Scattered by Design: Why Per-Output Pruning Resists Structured Compression

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Abstract

Network pruning for large language models operates in two regimes that have resisted unification. Unstructured methods such as Wanda preserve perplexity but yield no practical compression, while structured methods such as SliceGPT enable efficient inference at the cost of quality. We ask whether post-hoc linear transformations can convert the sparsity produced by per-output pruning into structured patterns. Across 13 fixed transformations and a learned cross-layer rotation, applied to OPT and LLaMA at 50% sparsity under three per-output pruning methods, we find no practically relevant increase in structured sparsity ($\Delta \leq 0.4\%$); the learned rotation is indistinguishable from a random one. We attribute this to rank preservation, spectral dispersion ($R_{10} < 1.0$), and weak inter-row mask correlation ($\rho(M) \approx 0.07$), properties contingent on the activation statistics the pruning score consumes rather than consequences of per-output pruning itself. A controlled synthetic study shows structure is recovered only when mask correlation forces rank collapse, and the one real violation we find (early-block attention output projections, $\Delta = +15.1\%$) is quality-neutral to delete yet yields no net compression. Rank-drop and spectral energy ratio serve as lightweight diagnostics, and we conclude that effective hybrid methods must introduce structure during pruning rather than recover it afterward.

1 Introduction

Large language models (LLMs) achieve strong performance by scaling in size [Kaplan et al., 2020], but their deployment is constrained by memory and compute requirements. Model compression is therefore essential, with network pruning offering a promising approach by removing weights from trained models. However, the effectiveness of pruning depends on the *structure* of the resulting sparsity. Unstructured pruning preserves model quality but yields irregular sparsity that cannot be exploited by standard hardware [Blalock et al., 2020], while structured pruning An et al. [2024], Ashkboos et al. [2024a], Ma et al. [2023] enables efficient inference but typically degrades performance. Bridging this gap remains an open challenge.

Recent methods highlight this tension. Wanda [Sun et al., 2024] achieves state-of-the-art perplexity at 50% sparsity by pruning weights based on joint weight activation importance[2], but produces fully unstructured sparsity. SliceGPT [Ashkboos et al., 2024a] applies orthogonal transformations followed by dimension removal to achieve real compression, but incurs noticeable perplexity degradation relative to wanda at equal compression rate. This contrast raises a natural question: can a post-hoc linear transformation reorganize Wanda, SparseGPT, and per-row magnitude's sparsity into structured patterns, combining both benefits?

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[2]candidate neuron for pruning are contigent on xW and not W alone.

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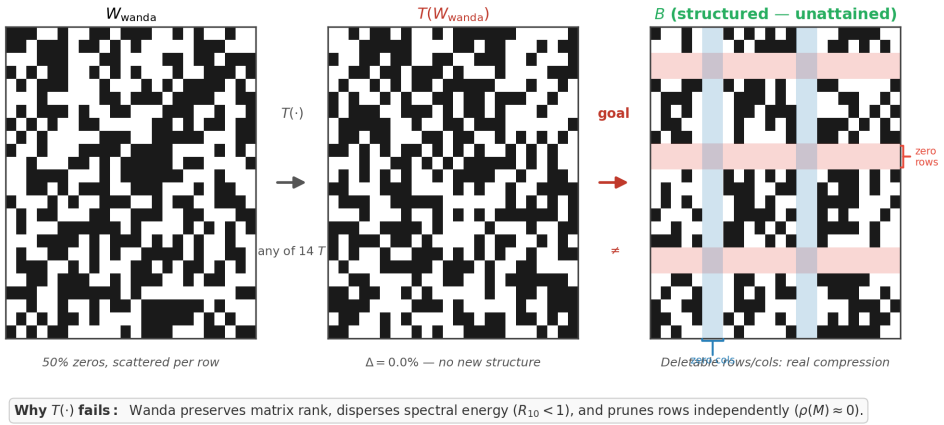


Figure 1: **Can a linear transformation map A to B?** Matrix A exhibits unstructured sparsity (e.g., Wanda-pruned weights), while B represents structured sparsity enabling row deletion. We ask whether there exists a transformation T such that $T(A)$ exhibits structure like B. **Answer: No.** Across a broad class of transformations, we observe no increase in structured sparsity. **Why:** A preserves rank, disperses spectral energy, and exhibits near-independent row sparsity, leaving no aligned structure for any linear map to exploit.

We show that this question can be answered negatively. Across a broad class of fixed, data-independent linear transformations (rotations, decompositions, statistical transforms, and permutations) together with a learned, data-dependent cross-layer rotation derived from activation covariance in the style of SliceGPT, applied to weights pruned by Wanda, SparseGPT, and per-row magnitude in OPT [Zhang et al., 2022] and LLaMA [Touvron et al., 2023] models, we observe no practically relevant increase in structured sparsity under any transformation ($\Delta \leq 0.4\%$). This holds across all layer types, model scales, sparsity metrics, and thresholds, other than the out projections of attention blocks.

We are able to explain this through three mechanisms. Per-output pruning preserves matrix rank, leaving no degenerate directions for transformations to collapse. It disperses rather than concentrates spectral energy, limiting low-rank structure. It further produces near-independent row masks, preventing alignment of sparsity across rows. Together, these properties remove the conditions under which post-hoc linear transformations could induce structured sparsity. Crucially, these are not consequences of the per-output rule itself but are contingent on the activation statistics the pruning score consumes. Via a controlled synthetic study, we present supporting evidence that structure emerges only when mask correlation is strong enough to force rank collapse. An exception is the attention output projection in an early layer where dead heads make those statistics degenerate, accounting for the single exception noted above.

Contributions. Our contributions are as follows:

- **A controlled negative result on post-hoc sparsity restructuring.** We evaluate 13 linear transformations (fixed data-independent maps) plus a learned cross-layer rotation on OPT and LLaMA models pruned by Wanda, SparseGPT, and per-row magnitude, and find no practically relevant increase in structured sparsity ($\Delta \leq 0.4\%$ across all settings).
- **A mechanistic explanation of transformation failure.** We identify three responsible conditions namely, rank preservation, spectral dispersion, and weak inter-row mask correlation. We show synthetically that structure emerges only when they are violated together, and trace the single exception (early-block attention output projections) to degenerate per-head activation statistics.
- **Predictive diagnostics for post-hoc compressibility.** We show that rank-drop consistency and spectral energy concentration serve as lightweight forward-pass diagnostics, validated in both negative and positive regimes.
- **A design principle for hybrid pruning methods.** We show that structure must be introduced during pruning rather than recovered post hoc, implying the need for structured importance scoring or jointly learned transformations.

- **A reproducible benchmark.** We release a codebase for standardized evaluation of pruning–transformation pipelines, built on the official Wanda and SparseGPT implementations.

2 Background

2.1 Unstructured Pruning

Network pruning removes weights from a trained model to reduce memory and computation. The metric for neuron removal can be based on magnitude, gradient, lottery ticket hypothesis [LeCun et al., 1989, Hassibi et al., 1993, Frankle and Carbin, 2019]. Generally, pruning removes weights with the smallest absolute values based on the pruning metric [Li et al., 2017]. Specifically, magnitude pruning performs poorly in large language models (LLMs) in relation to Wanda [Sun et al., 2024]. Wanda incorporates activation statistics into the pruning criterion. For a weight matrix $W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$ and input activations X, Wanda defines

$$S_{ij} = |W_{ij}| \cdot \|X_j\|_2,$$

where $\|X_j\|_2$ is the ℓ_2-norm of the j-th input feature estimated from a calibration set. Weights are pruned *per output neuron*, removing a fixed fraction within each row. This per-output comparison group preserves important connections for each neuron and achieves strong perplexity at moderate sparsity without retraining. However, the resulting sparsity is fully unstructured, that is, zeros are scattered across each row, yielding no reduction in parameter count or inference cost on standard hardware.

2.2 Structured Pruning

Structured pruning removes entire rows, columns, or dimensions to enable efficient computation with dense kernels [Li et al., 2017, Ma et al., 2023]. Unlike unstructured pruning, this yields real reductions in parameter count and FLOPs using standard hardware. However, at comparable sparsity, structured methods typically incur substantially larger degradation in model quality because removing entire components affects all downstream computations simultaneously. A typical structured pruner is SliceGPT [Ashkboos et al., 2024a]. It removes low-important dimensions of a residual stream after transformation. In practice, however, such dimension removal introduces noticeable perplexity degradation relative to unstructured pruning at comparable sparsity.

2.3 Compression Trade-off and Transformations

These approaches reflect a fundamental trade-off: unstructured pruning preserves model quality but produces irregular sparsity that is difficult to exploit, while structured pruning enables efficient inference but degrades performance. Intermediate schemes such as N:M sparsity [Mishra et al., 2021] impose partial structure, but do not fully resolve this gap.

A natural question is whether linear transformations can bridge these regimes. Transformations have been used in related settings such as quantization [Chee et al., 2023, Ashkboos et al., 2024b, Tseng et al., 2024]. It was used to redistribute statistical structure, thereby, improving compressibility. This suggests the possibility that a suitable transformation could reorganize the unstructured sparsity produced by methods like Wanda into structured patterns.

Our work examines this hypothesis directly, focusing on whether post-hoc linear transformations, fixed or derived from activation statistics, applied per-layer or across layers can induce structured sparsity in pruned LLM weights.

3 Related Work

3.1 LLM Pruning

Recent work has adapted classical pruning methods to large language models. SparseGPT [Frantar and Alistarh, 2023] formulates pruning as a layer-wise weight reconstruction problem using second-order approximations, achieving strong perplexity at moderate sparsity but requiring costly weight updates. Wanda [Sun et al., 2024] simplifies this by using a joint weight–activation importance

score, matching SparseGPT performance without retraining. Subsequent works explore structured or semi-structured pruning of LLM components such as attention heads and MLP neurons [Ma et al., 2023, Xia et al., 2023].

Distinction. These methods focus on *how to prune* while preserving model quality. In contrast, our work studies whether the resulting unstructured sparsity can be *post-processed* into structured form via linear transformations. We show that for Wanda-style pruning at moderate sparsity, this is not possible under a broad class of transformations, and provide a mechanistic explanation. It is important to note that learned transformation are excluded from our experiments because we want to keep the post-pruning finetuning-free advantage of wanda.

3.2 Structured and Transformation-Based Compression

Structured pruning methods remove entire rows, columns, or dimensions to enable hardware-efficient inference [Liu et al., 2017]. SliceGPT [Ashkboos et al., 2024a] extends this idea to LLMs by applying orthogonal transformations before dimension removal, leveraging a computational invariance property to preserve model outputs in the dense setting. FLAP [An et al., 2024] attempts structured pruning without rotation, operating at the column level — the approach our paper shows is fundamentally limited for Wanda-style metrics. LLM Surgeon [van der Ouderaa et al., 2024] a general framework for unstructured, semi-structured, and structured pruning of LLMs using block-diagonal Kronecker-factorized approximations to the empirical Fisher, deriving OBS-like weight updates for pruning multiple structured rows and columns jointly.

A related line of work uses linear transformations to improve compressibility in other regimes. QuIP and QuIP# [Chee et al., 2023, Tseng et al., 2024] apply random rotations to improve quantization, while SmoothQuant [Xiao et al., 2023] and QuaRot [Ashkboos et al., 2024b] redistribute scale between weights and activations to stabilize quantization. These methods demonstrate that transformations can meaningfully reshape weight and activation distributions. Furthermore, SpinQuant Liu et al. [2025] learns rotations on the Stiefel manifold for quantization, showing data-driven rotations outperform fixed transforms — we test the same approach for pruning and find no benefit (Table A2). SVD-LLM Wang et al. [2025] compresses LLMs via truncated SVD with activation-aware whitening; our $R_{10} < 1.0$ finding suggests Wanda-pruned matrices resist such low-rank compression more than their dense counterparts.

Distinction. Prior transformation-based methods either (i) operate jointly with the compression objective (e.g., before quantization or structured pruning), or (ii) assume dense weights. Our work instead studies *post-hoc transformations applied to already pruned weights*, asking whether they can reorganize binary sparsity patterns. We show that, unlike in quantization, such transformations fail to induce structure in unstructured sparsity, and identify the conditions responsible.

3.3 Sparsity Structure and Hardware Efficiency

The distinction between unstructured and structured sparsity has long been recognized [Blalock et al., 2020]. Unstructured sparsity offers flexibility but requires specialized sparse kernels, while structured sparsity enables efficient execution on standard hardware. Semi-structured approaches such as N:M sparsity [Mishra et al., 2021] impose local constraints (e.g., 2:4 patterns) to balance performance and efficiency.

Distinction. These works focus on *designing sparsity patterns* that are hardware-friendly. In contrast, we investigate whether such structure can emerge *after the fact* from fully unstructured sparsity. Our results show that element-wise pruning at moderate sparsity produces masks that lack the cross-row alignment required for transformation-induced structure, providing a negative answer to this possibility.

3.4 Post-Training Compression with Limited Data

Many modern compression methods operate in a post-training setting with limited calibration data, including SparseGPT, Wanda, and various quantization approaches [Frantar and Alistarh, 2023, Sun et al., 2024]. These methods rely on activation statistics estimated from small datasets to guide compression decisions.

Table 1: The 13 linear transformations evaluated, organised by family, together with the learned cross-layer rotations evaluated separately. †These produce zero rows on *any* matrix as a consequence of shape or decomposition structure, independent of pruning. ‡Derived from the residual-stream activation covariance of the *pruned* model and applied across layers by computational invariance; the Haar-random variant serves as a control.

Family	**Transforms**	**Acts on**	**Orthogonal?**
Orthogonal rot.	PCA, Hadamard, DCT	Input (right)	Yes
Matrix decomp.	SVD (left, right, two-sided †), QR†, LU	Both	Partially
Statistical	Diagonal scaling, SVD weighted	Input	No
Permutation	Row sort, Column sort, Cuthill-McKee	Both	Yes (discrete)
Learned, cross-layer‡	Global Q, per-block Q_i, Haar-random Q	Stream (both sides)	Yes

Distinction. Our work builds on this setting but shifts the focus from compression itself to *predicting compressibility*. We introduce simple diagnostics which are based on rank and spectral properties. They jointly determine whether a pruned model is amenable to structured compression without requiring downstream evaluation. This provides a complementary perspective to existing post-training methods.

4 Experiments

4.1 Setup

We evaluate post-hoc linear transformations on pruned weights across OPT [Zhang et al., 2022] (125M–6.7B) and LLaMA [Touvron et al., 2023] (7B, 13B), covering all linear layer types except embeddings and the language-modelling head. We prune to 50% unstructured sparsity using 128 C4 calibration sequences, with three per-output methods: Wanda [Sun et al., 2024], SparseGPT [Frantar and Alistarh, 2023] (including its second-order weight update), and per-row magnitude. The main sweep across scales prunes each analysed layer independently, with calibration statistics from the dense model. We repeat the analysis under the standard whole-model protocol, where pruning proceeds sequentially and each block is pruned on activations propagated through the already-pruned prefix [3]. These runs cover OPT-125M, with all three methods over three calibration draws, and LLaMA-7B, where our pruned model reproduces the published WikiText-2 perplexity (7.257 vs. 7.26). Conclusions are unchanged under both protocols. We additionally sweep sparsity over 50–80% on OPT-1.3B.

To enable cross-model comparison on a common axis, we stratify each model into 10 depth bins and sample the central block from each bin, yielding 10 representative blocks regardless of total depth. For each pruned matrix W_s, we measure the *marginal contribution* of pruning's zeros relative to the same transform on the dense matrix:

$$\Delta(T, W_s) = \text{zero_rows}(T(W_s)) - \text{zero_rows}(T(W)). \tag{1}$$

$\Delta \approx 0$ means the transform produces no structure attributable to pruning. We report Δ across four sparsity metrics (exact row sparsity, near-zero rows, 4×4 block sparsity, column sparsity) and four thresholds (10^{-3}–10^{-6}); Table 2 reports exact row sparsity at 10^{-6} as the primary metric. Results are stable across all metrics and thresholds (see Table A1 in the Appendix).

4.2 Transformations Evaluated

We evaluate 13 fixed, data-independent transformations (Table 1) spanning orthogonal rotations (PCA, Hadamard, DCT), matrix decompositions (SVD in its left, right, and two-sided forms, QR, LU), statistical methods (diagonal scaling, SVD weighted), and permutations (row sort, column sort, Cuthill-McKee). The identity map is included throughout as a baseline. We additionally evaluate a learned, data-dependent rotation in the style of SliceGPT [Ashkboos et al., 2024a]. It was derived

[3] We used the official Wanda and SparseGPT implementations

from the residual-stream activation covariance of the pruned model and applied across layers by computational invariance. We test a single global Q, a per-block Q_i, and a Haar-random Q as a control; full details are in Appendix A. Jointly-optimised transforms remain excluded by design. Optimising a transform against a pruning objective requires gradient computation and post-pruning fine-tuning, which defeats Wanda's key advantage of requiring no weight update.

Table 2: Marginal contribution of pruning: $\Delta = \text{zero_rows}(T(W_{\text{wanda}})) - \text{zero_rows}(T(W_{\text{dense}}))$. Values are from the per-layer protocol; where whole-model sequential pruning was also run, those values appear in brackets. Whole-model OPT-125M values are means over three independent calibration draws (std $\leq 1.3 \times 10^{-4}$ for every transform); LLaMA-7B uses a single draw. Non-zero entries are driven by the single `out_proj`/`o_proj` anomaly layer (Table 6).

Transform	**OPT-125M**	**OPT-1.3B**	**OPT-6.7B**	**LLaMA-7B**	**LLaMA-13B**
PCA (right)	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%
Hadamard (right)	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%
DCT (right)	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%
SVD right (V)	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%
SVD left ($U^\top$)	+0.1% (+0.33%)	0.0%	0.0%	+0.1% (+0.22%)	0.0%
SVD two-sided $(\Sigma)^\dagger$	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%
QR left ($\rightarrow R)^\dagger$	0.0% (+0.02%)	0.0%	0.0%	0.0% (0.00%)	0.0%
LU left ($\rightarrow U$)	0.0% (+0.03%)	0.0%	0.0%	0.0% (0.00%)	0.0%
Diagonal scaling	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%
SVD weighted	+0.2% (+0.33%)	0.0%	0.0%	+0.1% (+0.22%)	0.0%
Column sort	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%
Row sort	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%
Cuthill-McKee	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%
Identity (baseline)	0.0% (0.00%)	0.0%	0.0%	0.0% (0.00%)	0.0%

5 Main Result

Fixed transformations. Table 2 reports Δ across the OPT and LLaMA model families. *No transformation produces practically relevant structured sparsity attributable to pruning.* The largest value anywhere is $+0.33\%$, two orders of magnitude below the range at which structured methods deliver measurable efficiency gains. This holds on FFN layers, on attention projections other than `out_proj`/`o_proj`, and at every threshold from 10^{-3} to 10^{-6}.

Transforms that do produce zero rows, namely QR (45–75%) and SVD (100%), produce identical structure on the dense matrix shown in Table 4. Their Δ is therefore near zero, confirming that the structure reflects intrinsic matrix geometry rather than the placement of pruned weights.

Table 3: Generality across per-output pruning methods: $\Delta = \text{zero_rows}(T(W_{\text{pruned}})) - \text{zero_rows}(T(W_{\text{dense}}))$ on OPT-125M at 50% sparsity under whole-model pruning. Three representative transforms are shown: the identity baseline, the transform attaining the largest Δ, and one orthogonal rotation standing for the remaining ten. The final row is the maximum over all 13 transforms. Full per-transform results are in Appendix B.

Transform	**Wanda**	**SparseGPT**	**Magnitude (per-row)**
Identity (baseline)	0.00%	+0.20%	0.00%
SVD left ($U^\top$)	+0.33%	+0.06%	+0.03%
Hadamard (right)	0.00%	+0.01%	0.00%
Maximum over 13 transforms	**+0.33%**	**+0.20%**	**+0.04%**

Generality across pruning methods. The result is not specific to Wanda. Table 3 applies the same 13 transformations to OPT-125M pruned by three per-output methods that differ in what information they consume. Wanda scores by magnitude and activation norm. SparseGPT adds a Hessian-based

weight update, the mechanism most likely to alter spectral structure. Per-row magnitude consults no activation statistics at all. Maximum Δ is $+0.33\%$, $+0.20\%$ and $+0.04\%$ respectively, and the method sharing only the per-output comparison group gives the cleanest null.

Table 4: Zero-row fraction of QR and SVD two-sided applied to *dense* unpruned matrices. Values are identical on Wanda-pruned matrices ($\Delta = 0.0\%$), confirming these structures are intrinsic to matrix geometry. QR: $\max(0, C_{\text{out}} - C_{\text{in}})/C_{\text{out}}$ (shape artifact); SVD two-sided: 100% by construction.

Transform (dense W)	**OPT-125M**	**OPT-1.3B**	**OPT-6.7B**	**LLaMA-7B**	**LLaMA-13B**
QR left ($\rightarrow R$)	13.1%	13.2%	13.2%	18.6%	18.7%
SVD two-sided (Σ)	100.0%	100.0%	100.0%	100.0%	100.0%

SparseGPT's largest value occurs under the *identity* map, recurring under every transform that preserves exact zeros. These are output rows the weight update zeroed during pruning, present before any transformation and improved by none.

5.1 Learned Cross-Layer Rotations

The transformations above are fixed and act on one layer at a time. A natural objection is that a rotation derived from the model's own activations, applied consistently across layers, might succeed where fixed maps fail. This is the class SliceGPT [Ashkboos et al., 2024a] operates in. We therefore compute an orthogonal Q from the residual-stream activation covariance of the *pruned* model and apply it across layers by computational invariance (Appendix A). Because slicing removes small-norm rather than exactly zero slices, we report near-zero metrics alongside the exact ones.

Table 5: Learned cross-layer rotations on whole-model-pruned OPT-125M at 50% sparsity. Δ is measured against the same rotation applied to the dense weights. The covariance-derived rotations are indistinguishable from a Haar-random control on every metric. Column-side entries are identical across all three variants.

Pruning	**Rotation**	**Δ zero rows**	**Δ near-zero rows**	**Δ near-zero cols**
Wanda	Global Q	0.00%	0.00%	+1.25%
	Per-block Q_i	0.00%	0.00%	+1.25%
	Haar-random Q	0.00%	0.00%	+1.25%
SparseGPT	Global Q	0.00%	0.00%	+0.98%
	Per-block Q_i	0.00%	0.00%	+0.99%
	Haar-random Q	0.00%	0.00%	+0.98%

No cross-layer rotation produces a single additional deletable row, for either pruning method. More tellingly, the rotation computed from activation statistics performs exactly as well as a random orthogonal one. The failure is therefore not a matter of choosing Q poorly. There is no basis in which the pruned mass concentrates into rows, so no orthogonal cross-layer map can find one. The only non-zero entries are column-side, identical across all three variants, and attributable to pruning itself: entire columns of dead intermediate neurons, which orthogonal maps preserve but cannot create.

5.2 The Output-Projection Anomaly

A single exception emerges across OPT and LLaMA. The output projection of early attention blocks exhibits a genuine rank drop under Wanda pruning (Table 6). The strongest case is LLaMA-7B `layers.1.o_proj`, where SVD-left attains $\Delta = +15.1\%$, the largest value observed across all models and transforms. Under whole-model pruning the anomaly extends to the preceding block in both OPT-125M and LLaMA-7B (Appendix C).

The anomaly is real but yields no usable compression. We rank-truncate that layer to exactly the rank its SVD-left zero rows imply, deleting 618 of 4096 rows. WikiText-2 perplexity is unchanged ($7.2568 \rightarrow 7.2570$), so the deleted directions carry no information. Factored storage of the truncated layer costs 28.5M parameters against 16.8M dense, however, since factorisation only reduces storage below rank $n/2$ and the retained rank is 3478. Even the strongest violation we observe therefore yields no net compression. We explain its cause in Section 6.5.

Table 6: `out_proj`/`o_proj` layers with genuine rank-drop > 10. Values are from the per-layer protocol, with whole-model sequential pruning in brackets. Spectral and effective-rank diagnostics are in Appendix C.

Model	**Layer**	**rank-drop**		$\rho(M)$	Δ
		10^{-3}	10^{-6}		**(SVD-left)**
OPT-125M	layers.1.out_proj	+67 (+85)	+66 (+84)	0.415 (0.414)	+8.6% (+10.9%)
OPT-1.3B	layers.3.out_proj	+50	+37	0.397	+1.8%
OPT-6.7B	layers.17.out_proj	+52	+58	0.146	+1.4%
LLaMA-7B	layers.1.o_proj	+428 (+591)	+431 (+618)	0.460 (0.513)	+10.5% (+15.1%)
LLaMA-13B	layers.2.o_proj	+13	+0	0.157	+0.0%

Table 7: Diagnostic metrics. *rank-drop* $= \mathrm{rank}(W) - \mathrm{rank}(W_{\text{wanda}})$ at 10^{-3} and 10^{-6}. ***hi90*** = rank-drop at 90% magnitude sparsity (positive control). R_{10} = top-10 SV energy ratio. $\rho(M)$ = inter-row mask correlation. Sign reversal across thresholds indicates spectral redistribution. R_{10} and $\rho(M)$ report mean $\pm$ std across sampled layers.

Model	**rank-drop** 10^{-3}	**rank-drop** 10^{-6}	**hi90**	R_{10} (mean $\pm$ std)	$\rho(M)$ (mean $\pm$ std)
OPT-125M	-2.25	$+1.10$	$+13.52$	0.945 ± 0.023	0.073 ± 0.114
OPT-1.3B	-6.65	$+0.60$	$+5.80$	0.939 ± 0.019	0.068 ± 0.098
OPT-6.7B	-3.80	$+0.97$	$+15.42$	0.934 ± 0.017	0.060 ± 0.090
LLaMA-7B	$+0.49$	$+6.11$	$+52.34$	0.939 ± 0.009	0.032 ± 0.057
LLaMA-13B	-4.46	-0.03	$+36.49$	0.939 ± 0.006	0.026 ± 0.030

5.3 Diagnostic Metrics

Table 7 reports three diagnostic metrics across model families. Three patterns are consistent:

(1) Rank-neutrality. Mean rank-drop ranges from -6.65 (OPT-1.3B) to $+0.49$ (LLaMA-7B) at threshold 10^{-3}, reversing to near-zero at 10^{-6} (-0.03 to $+6.11$). We consider Sign reversal across thresholds to be a signature of spectral redistribution rather than rank collapse: a genuine rank drop would manifest as a consistent nonzero value across all thresholds, whereas threshold-sensitive fluctuations reflect the well-known sensitivity of numerical rank to the choice of tolerance for matrices with smoothly decaying spectra [Golub and Van Loan, 2013, Trefethen and Bau, 1997].

The same pattern holds for magnitude pruning at 50% sparsity, suggesting this behaviour is a general consequence of element-level sparsification at moderate density rather than a property of Wanda's scoring criterion. At 90% sparsity, both methods produce consistent rank drops of $+5.8$ to $+52.3$ across all thresholds, validating the diagnostic in the positive regime.

(2) Spectral dispersion. R_{10} (top-10 SV energy ratio, $W_{\text{wanda}}/W_{\text{dense}}$) is consistently 0.93–0.95 across all models and depths, meaning Wanda *reduces* top-singular-value concentration relative to the dense matrix.

(3) Mask independence. Mean inter-row mask correlation $\rho(M)$ ranges from 0.032 (LLaMA-7B) to 0.073 (OPT-125M), compared to 0.011–0.026 for a random i.i.d. mask of identical sparsity. Wanda's masks are nearly as uncorrelated as random, leaving no cross-row alignment for any linear transform to exploit.

6 Why Post-Hoc Transformation Fails: Three Mechanisms

Our experiments establish $\Delta \leq 0.4\%$ across all 13 transforms, all layer types, and all thresholds. Three properties of pruned matrices, acting jointly, explain this. We show below that they are not consequences of the per-output pruning rule but are contingent on the activation statistics that rule consumes, which is why they can be measured, and occasionally violated.

6.1 Rank Preservation

For any transformation T to produce zero rows in $T(W_s)$, the pruned matrix must be rank-deficient. We compute rank-drop $= \text{rank}(W) - \text{rank}(W_s)$ at $\tau \in \{10^{-3}, \ldots, 10^{-6}\}$, all anchored to $\sigma_{\text{max}}(W)$ to prevent spurious differences across matrices.

FFN layers show rank-drop $= 0$ at all thresholds across all models. Attention projections show small sign-inconsistent fluctuations, for example -6.65 to $+0.60$ in OPT-1.3B as τ tightens. A genuine rank drop is consistent across thresholds, not threshold-dependent [Golub and Van Loan, 2013]; 90% magnitude pruning yields $+5.8$ to $+52.3$, positive at all thresholds (Table 7). The cause is structural. Pruning exactly $s\%$ of weights *within* each row leaves every row with $(1-s)$ of its weights, which at $s = 0.5$ suffices with high probability to maintain linear independence.

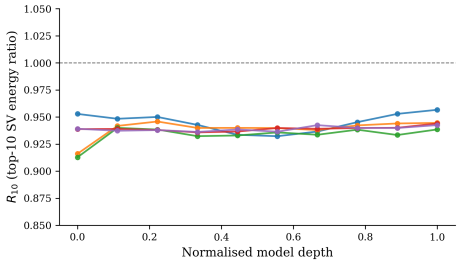


Figure 2: R_{10} (top-10 SV energy ratio, $W_{\text{wanda}}/W_{\text{dense}}$) across normalised depth. All curves lie below $R_{10} = 1.0$ (dashed): Wanda disperses spectral energy, ruling out low-rank post-hoc compression across all scales (125M–13B).

6.2 Spectral Dispersion

Even at full rank, rapidly decaying singular values admit accurate low-rank approximation [Eckart and Young, 1936]. We measure the top-k energy ratio $R_k = \sum_{i \le k} \sigma_i(W_s) / \sum_{i \le k} \sigma_i(W)$, where $R_k > 1$ indicates concentration. We find $R_{10} \in [0.93, 0.95]$ across all models and depth bins (Fig. 2). Pruning *disperses* spectral energy: weights connected to high-activation inputs are broadly distributed, so retaining them while zeroing the rest flattens the spectrum. Pruned matrices are therefore harder to compress post-hoc than their dense counterparts.

6.3 Mask Independence

For T to align zeros into rows, the mask must exhibit cross-row correlation. We measure the mean absolute Pearson correlation across row pairs, $\rho(M)$. Across all models $\rho(M) = 0.032$–0.073, against 0.011–0.026 for a random i.i.d. mask of identical sparsity (Table 7). Rows are pruned nearly independently because the per-output score $S_{ij} = |W_{ij}| \cdot \|X_j\|_2$ depends on row i only through W_{ij}: shared activation norms weight columns uniformly but do not correlate *which* columns are pruned across rows.

6.4 Causal Validation

The three properties co-occur in real models, so observational data cannot separate them. We therefore construct synthetic matrices ($n = 512$, 50% sparsity, 5 seeds) in which spectrum, rank, and mask correlation are varied independently (Appendix D). The results isolate the operative condition. With independent masks ($\rho = 0.035$), $\Delta \le 0$ in *every* spectrum and rank configuration: violating rank or spectrum alone recovers nothing. With correlated masks but preserved rank ($\rho = 0.16$), $\Delta = +0.1\%$. Only when correlation is strong enough to force rank collapse does structure appear, and then Δ equals the imposed violation exactly ($\rho = 0.53$, rank-drop 125, $\Delta = +24\%$; shared-column fraction c gives $\Delta \approx c$). Mask correlation is thus the entry condition and rank collapse the operative gate, with dispersion controlling whether near-zero rows survive at practical thresholds. Failure under per-output pruning is overdetermined. Mask independence alone blocks recovery, and rank preservation blocks it even under correlation.

6.5 The Anomaly Explained

This account predicts where violations should occur, and it locates them. Unlike other projections, `out_proj` reads the concatenated outputs of the attention heads. Its input features are therefore grouped, making activation norms vary far more between heads than within them. We quantify this as the share of variance in the pruning score explained by head identity, which reaches 0.92

at `out_proj` and stays below 0.01 at `q_proj`, which has identical shape at every depth. Grouping alone changes nothing. The trigger is *dead heads*, whose value outputs are small enough that the per-row threshold prunes their entire column block in every row. In LLaMA-7B, 5 of 32 and 10 of 32 heads are dead in blocks 0 and 1, and none thereafter, tracking the anomaly exactly while across-head heterogeneity continues to rise with depth (Appendix E). Correlation without rank consequence again yields nothing, as in the synthetic study. The violation is also Wanda-specific. SparseGPT prunes the same positions but its weight update writes compensation values back, reducing the mean out_proj rank-drop from $+12.0$ to $+0.5$ and Δ from $+1.5\%$ to $+0.07\%$ on OPT-125M.

6.6 Diagnostic Criteria

The mechanisms yield two forward-pass diagnostics. **Rank-drop consistency**: positive rank-drop at all τ predicts compressibility; sign-inconsistent or near-zero rank-drop predicts failure. **Spectral energy ratio**: $R_{10} > 1.0$ is necessary for effective low-rank post-hoc compression. We validate both positively with 90% magnitude pruning (rank-drop $+5.8$ to $+52.3$, $\Delta > 0$ under SVD) and negatively with per-output pruning at 50% ($R_{10} = 0.93$–0.95, $\Delta \leq 0.4\%$). Both require one calibration forward pass and two SVDs, far cheaper than running the full benchmark.

7 Discussion

Where a viable hybrid must intervene. The three mechanisms of Section 6 share a common origin in the per-output comparison group, which evaluates each output neuron independently. They are not, however, automatic consequences of that design. They hold when the activation statistics the score consumes are roughly homogeneous across input features, and they fail when those statistics are degenerate, as at early-block output projections. This is why the properties must be measured rather than assumed, and why the diagnostics of Section 6.6 carry information. What follows for method design is unchanged: across three per-output methods, spanning mask-only scoring, second-order weight updates, and no activation information at all, no post-hoc transformation recovers usable structure. A viable hybrid must introduce structural bias before or during pruning. We conjecture that replacing the per-output comparison group with column-spanning groups, analogous to SparseGPT's 128-channel grouping would address both rank preservation and mask independence, and leave empirical validation to future work.

Structure originates at pruning time. Our whole-model experiments make this concrete. The only exact structured sparsity we observe anywhere is created by pruning itself: Wanda and SparseGPT independently zero the outgoing columns of the same dead intermediate neurons, reaching 40.8% of columns in one early FFN layer. Orthogonal maps preserve column norms on the write side, so this structure is invariant to any such transform, and no transform in our suite improves on it.

Broader implications. Our results reframe the structured/unstructured trade-off. The traditional view treats structure as a property of the pruning *output*, the pattern of zeros. Our analysis shows it must be a property of the pruning *objective* built into the comparison group or the basis in which importance is evaluated. Post-hoc transformation attempts to recover cross-row relationships that were never captured during pruning. The path to bridging the two regimes runs through the importance scoring function, not around it.

8 Conclusion.

We asked whether post-hoc linear transformations can convert the scattered zeros of per-output pruning into structured patterns enabling real compression. The answer is no. Across 13 fixed transforms and a learned cross-layer rotation, three pruning methods, and scales from 125M to 13B, $\Delta \leq 0.4\%$ at every threshold. The single exception, early-block attention output projections ($\Delta = +15.1\%$), confirms rather than contradicts the finding: deleting its rows leaves perplexity unchanged but yields no net compression. Three mechanisms explain the failure: rank preservation, spectral dispersion, and mask independence, each contingent on the activation statistics the pruning score consumes. We introduce rank-drop consistency and spectral energy ratio as lightweight forward-pass diagnostics, validated in positive and negative regimes.

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Appendix

Table A1: Wanda's marginal contribution Δ across all four sparsity metrics and all thresholds, averaged across transforms and layers. All entries are $\leq 0.1\%$, confirming that $\Delta \approx 0$ is not specific to the exact-row metric or any particular threshold. *Exact*: fraction of fully-zero rows. *Near-zero*: fraction of rows with norm $< 1\%$ of mean row norm. *Block*: fraction of 4×4 blocks that are fully zero. *Column*: fraction of fully-zero columns. Values are mean Δ (wanda $-$ dense) across all 14 transforms(include the identity baseline) and all sampled layers per model.

Metric	**Threshold**	**OPT-125M**	**OPT-1.3B**	**OPT-6.7B**	**LLaMA-7B**	**LLaMA-13B**
Exact rows	10^{-3}	+0.19%	+0.23%	+0.28%	+0.17%	+0.12%
	10^{-4}	+0.04%	+0.03%	+0.03%	+0.04%	+0.01%
	10^{-6}	+0.02%	+0.00%	+0.00%	+0.02%	+0.00%
Near-zero	—	-0.11%	-0.04%	-0.00%	+0.01%	-0.02%
Block 4x4	—	+0.42%	+0.52%	+0.44%	+0.22%	+0.10%
Columns	—	+0.81%	+0.81%	+0.61%	+0.17%	+0.01%

Table A2: Pre-pruning rotation experiment on OPT-125M (10 largest layers). Row-importance entropy reduction under orthogonal Q applied *before* Wanda scoring. PCA rotation (activation-variance basis) and Stiefel manifold optimisation of Q both fail to concentrate row importance, with entropy reductions below 0.1% on average. This suggests that activation-variance-based rotation saturates quickly and joint optimisation of Q and the pruning mask is needed.

Method	**Description**	**Mean ΔH**	**Max ΔH**
Identity (no rotation)	Baseline: Wanda scores on W	0.00%	0.00%
PCA rotation	Right-multiply by activation PCA basis	0.06%	0.38%
Stiefel opt. (output)	Left-multiply, entropy minimised on Stiefel mfd.	0.04%	0.13%

ΔH = normalised entropy reduction of Wanda row-importance scores. Higher = more concentrated.

A Learned Cross-Layer Rotations

The transformations in the main suite are fixed and applied per layer. To test the strongest practical post-hoc alternative, we evaluate a data-dependent rotation in the style of SliceGPT [Ashkboos et al., 2024a].

After whole-model pruning, we accumulate the residual-stream activation covariance Σ of the *pruned* model over C4 calibration data. The transform therefore sees only what a post-hoc method would see. We take Q to be the eigenbasis of Σ, ordered by descending eigenvalue. Following computational invariance, Q is applied across layers rather than within one. Weights reading from the residual stream are mapped to WQ (`q/k/v_proj`, `fc1/up_proj`, `gate_proj`) and weights writing to it to $Q^\top W$ (`out_proj/o_proj`, `fc2/down_proj`).

We consider three variants: a single global Q pooled over all blocks, a per-block Q_i from that block's input covariance, and a Haar-random orthogonal Q as a control. Slicing removes small-norm slices rather than exactly zero ones, so we report Δ for near-zero rows and columns alongside the exact metrics.

We analyse transformed weights only and do not modify the model. The LayerNorm-to-RMSNorm absorption that exact functional invariance requires on OPT [Ashkboos et al., 2024a] is therefore not needed here.

B Per-Transform Results Across Pruning Methods

Table A3 gives the complete per-transform breakdown summarised in Table 3. Wanda and SparseGPT values are means over three independent calibration draws; per-row magnitude pruning is deterministic and uses a single run.

[Results table: Δ per variant, OPT-125M, Wanda and SparseGPT.]

C Output-Projection Anomaly: Full Diagnostics

Table A4 extends Table 6 with spectral and effective-rank diagnostics, and adds the two layers that reach the rank-drop criterion only under whole-model sequential pruning. These are the blocks immediately preceding the layers reported in the main text, so the anomaly spans the first two blocks of each model rather than a single layer.

D Synthetic Validation of the Three Mechanisms

Construction. We generate $W = U\mathrm{diag}(\sigma)V^\top$ with $n = 512$ and random orthogonal U, V, varying three factors independently. *Spectrum*: flat ($\sigma_i = 1$) or concentrated ($\sigma_i = e^{-8i/n}$). *Rank*: full, or half ($\sigma_i = 0$ for $i > n/2$). *Mask correlation*, all at 50% sparsity: independent per-row masks; G row-groups sharing one mask each, where large G gives correlation whose column support still covers every column; and a shared zero-column fraction c, which guarantees rank loss. We report Δ under SVD-left, averaged over 5 seeds, using the relative zero threshold $10^{-6} \max |T(W)|$.

Table A3: Complete per-transform Δ (exact zero rows) on OPT-125M at 50% sparsity under whole-model pruning. SparseGPT's elevated entries coincide exactly with the transforms that preserve exact zeros (identity, permutations, diagonal scaling), indicating rows zeroed by the weight update during pruning rather than structure recovered by any transform.

Transform	**Wanda**	**SparseGPT**	**Magnitude (per-row)**
PCA (right)	0.00%	+0.01%	0.00%
Hadamard (right)	0.00%	+0.01%	0.00%
DCT (right)	0.00%	+0.01%	0.00%
SVD right (V)	0.00%	+0.01%	0.00%
SVD left ($U^\top$)	+0.33%	+0.06%	+0.03%
SVD two-sided (Σ)	0.00%	0.00%	0.00%
QR left ($\rightarrow R$)	+0.02%	+0.01%	-0.00%
LU left ($\rightarrow U$)	+0.03%	+0.03%	0.00%
Diagonal scaling	0.00%	+0.20%	0.00%
SVD weighted	+0.33%	+0.06%	+0.04%
Column sort	0.00%	+0.20%	0.00%
Row sort	0.00%	+0.20%	0.00%
Cuthill-McKee	0.00%	+0.20%	0.00%
Identity (baseline)	0.00%	+0.20%	0.00%
Maximum	**+0.33%**	**+0.20%**	**+0.04%**

Table A4: Rank and spectral diagnostics for `out_proj`/`o_proj` layers with rank-drop > 10. Per-layer protocol, with whole-model sequential pruning in brackets. Rows marked † reach the criterion only under whole-model pruning. Transform results for the same layers are in Table A5.

Model	**Layer**	**rank-drop**		$\rho(M)$	R_{10}
		10^{-3}	10^{-6}		
OPT-125M†	l.0.out_proj	(+57)	(+58)	(0.311)	(0.984)
OPT-125M	l.1.out_proj	+67 (+85)	+66 (+84)	0.415 (0.414)	0.994 (0.994)
OPT-1.3B	l.3.out_proj	+50	+37	0.397	0.986
OPT-6.7B	l.17.out_proj	+52	+58	0.146	0.936
LLaMA-7B†	l.0.o_proj	(+142)	(+40)	(0.565)	(0.991)
LLaMA-7B	l.1.o_proj	+428 (+591)	+431 (+618)	0.460 (0.513)	0.985 (0.984)
LLaMA-13B	l.2.o_proj	+13	+0	0.157	0.953

Reading the table. Independent masks give $\Delta \leq 0$ in both spectrum settings, and also when the dense matrix is rank-deficient (not shown): violating rank or spectrum without correlation recovers nothing. Correlation alone is likewise insufficient. At $G{=}32$ and $G{=}8$ the masks are 2–4 times more correlated than the independent baseline, yet the column support still spans all columns, rank is preserved, and Δ stays at zero. Only at $G{=}2$, where the support shrinks and rank falls by 125, does structure appear. The shared-column rows make the correspondence exact: $\Delta \approx c$ for every c, so what is recovered is precisely what was imposed and nothing more. Spectral setting shifts Δ only slightly, indicating that dispersion governs whether near-zero rows survive at practical thresholds rather than whether exact zero rows exist. On half-rank dense matrices Δ is strongly negative, because the dense baseline itself already yields zero rows under SVD-left.

E Output-Projection Anomaly: Mechanism

Protocol. For every block we compare the target layer (`out_proj`/`o_proj`) against two controls under whole-model pruning: `q_proj`, which has identical shape at every depth, and the FFN write-side projection. We decompose the variance of the pruning score's activation term across versus within head-sized column groups, reporting the across-head share, and we count heads whose column block is more than half zeroed.

Table A5: Effective-rank change and marginal contribution under SVD-left for the layers of Table A4. Effective rank is the entropy of the normalised singular-value distribution; it was not recomputed under the whole-model protocol.

Model	**Layer**	**eff-rank change**	**Δ (SVD-left)**
OPT-125M†	l.0.out_proj	—	(+7.6%)
OPT-125M	l.1.out_proj	-42	+8.6% (+10.9%)
OPT-1.3B	l.3.out_proj	-94	+1.8%
OPT-6.7B	l.17.out_proj	-18	+1.4%
LLaMA-7B†	l.0.o_proj	—	(+1.0%)
LLaMA-7B	l.1.o_proj	-312	+10.5% (+15.1%)
LLaMA-13B	l.2.o_proj	-26	+0.0%

Table A6: Synthetic factorial, full-rank dense matrices. Structure is recovered only when mask correlation forces rank collapse, and then in the amount of the imposed violation. The G=8 row isolates correlation without rank consequence.

Mask	**Spectrum**	$\rho(M)$	**rank-drop**	R_{10}	Δ
Independent	flat	0.035	0.0	1.256	-0.2%
Independent	concentrated	0.035	-66.0	0.543	0.0%
Groups, G=32	flat	0.063	0.2	1.256	-0.2%
Groups, G=8	flat	0.156	1.6	1.246	$+0.1\%$
Groups, G=2	flat	0.527	125.0	1.144	$+24.2\%$
Groups, G=2	concentrated	0.527	56.0	0.664	$+24.4\%$
Shared cols, c=0.10	flat	0.111	51.0	1.270	$+9.8\%$
Shared cols, c=0.25	flat	0.333	128.0	1.281	$+24.8\%$
Shared cols, c=0.40	flat	0.662	204.0	1.257	$+39.6\%$

Enabling structure is not the trigger. Head identity explains 66% (LLaMA-7B) and 20% (OPT-125M) of the variance of the activation term at the output projection, against at most 1% at either control. Shape is excluded by construction, since `q_proj` is $d \times d$ at every depth. The `fc2` row is instructive: its $\rho(M)$ is the highest in the table, driven by dead intermediate neurons, yet its rank-drop is zero, because 768×3072 leaves far more live columns than rows. Correlation without rank consequence again produces nothing, exactly as in Table A6.

Dead heads are the trigger. Blocks 0 and 1 contain heads whose value outputs are small enough that the per-row threshold prunes their entire column block in every row. No block beyond the second contains such a head, and no block beyond the second shows the anomaly, despite across-head heterogeneity continuing to rise. Moderately weak heads at depth are compensated by weight magnitude; only near-zero heads are not. In block 1 this zeroes 23.6% of the layer's columns outright, which is the structure SVD-left then exposes as zero rows.

F Limitations

Our transform class covers fixed per-layer maps and a learned, cross-layer rotation derived from activation statistics. Transforms optimised *jointly* with a pruning objective are excluded, since training one requires gradients and post-pruning fine-tuning, forfeiting the advantage of update-free pruning; whether such a method could succeed remains open, and our mechanism suggests it is where the leverage lies. Our rank-preservation argument is specific to moderate sparsity. We verify it from 50% to 80%, but at $\geq 90\%$ rank does drop and post-hoc structured compression becomes viable (Table 7), though at that point model quality is typically compromised. Experiments cover OPT and LLaMA at six scales; architectures with different attention mechanisms, such as mixture-of-experts, may behave differently. Finally, we define structured sparsity as zero-row fraction and 4×4 block sparsity, and do not evaluate hardware-specific patterns such as 2:4 sparsity, which impose weaker structural requirements than full row deletion.

Table A7: Layer-type comparison under whole-model pruning. All values are means over every block of that layer type, so the target rows understate the anomaly: LLaMA-7B `o_proj` averages $+23.5$ rank-drop and $+0.50\%$ Δ across 32 blocks, of which only blocks 0 and 1 are affected (Table A8, worst block $+591$ and $+15.09\%$). The comparison across layer types remains valid, since the controls are averaged identically. Head-block structure in the activation statistics is present only at the output projection, and only there do the three properties fail.

Model	**Layer**	**Role**	**across-head**	$\rho(M)$	**rank-drop**	Δ
LLaMA-7B	`o_proj`	target	0.663	0.116	$+23.5$	$+0.50\%$
	`q_proj`	shape control	0.009	0.019	-27.1	$+0.01\%$
	`down_proj`	write control	0.003	0.036	0.0	0.00%
OPT-125M	`out_proj`	target	0.203	0.111	$+12.0$	$+1.54\%$
	`q_proj`	shape control	0.010	0.034	-9.3	-0.01%
	`fc2`	write control	0.003	0.210	0.0	0.00%

Table A8: Depth profile of LLaMA-7B `o_proj`. Dead heads, not head-block heterogeneity, track the anomaly: the across-head share rises with depth while the anomaly disappears after block 1.

Block	**across-head**	**dead heads**	$\rho(M)$	**rank-drop**	Δ
0	0.174	5/32	0.565	$+142$	$+0.98\%$
1	0.500	10/32	0.513	$+591$	$+15.09\%$
2	0.415	0/32	0.195	$+14$	$+0.02\%$
3–15 (mean)	0.585	0/32	0.067	-0.8	0.00%
16–31 (mean)	0.782	0/32	0.098	$+1.0$	0.00%

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Justification:

Justification: Table 7 reports mean $\pm$ std across sampled layers for R_{10} and $\rho(M)$. Table 2 notes that std of $\Delta < 0.001$ across layers, confirming $\Delta = 0.0\%$ is not an averaging artifact. The `out_proj` anomaly values in Table 6 are individual layer measurements, not aggregates.

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